题一：configuration (R or S)

判断：① 手性

② 从C连接的第一个原子看原子序数，谁越大排前，H最后

③ H是虚线时顺序不翻转：顺 R 逆 S，反之

题二：aromatic

特征：the criteria for aromaticity and Hückel's Rule：

① Cyclic structure

② Planarity

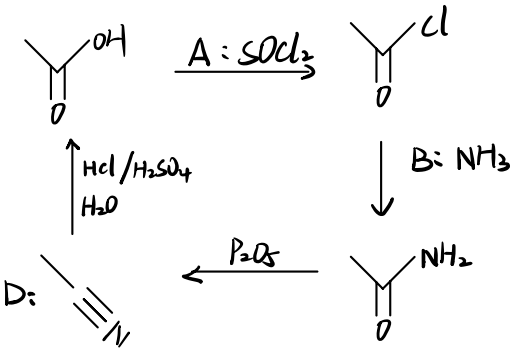
③ Conjugation (π electrons must be delocalized over the entire ring)

④ Follow Hückel's rule ($4n+2$ π electrons)

判断：双键 2个π，孤电子对 2个π，⊖ 2个π，⊕为0，须连续

例：  Aromatic 2π electrons, planar, every atom involved

题三：reagents A, B, C and product D



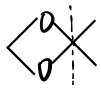
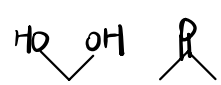
题四：increasing acidity

reason：① acidity：carboxylic acids > Phenol > Alcohol

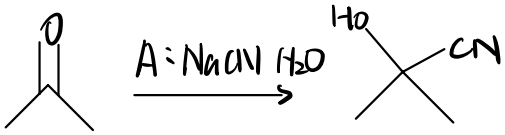
② Inductive effect：F (fluorine) withdraw electron density, stabilizing the conjugate base

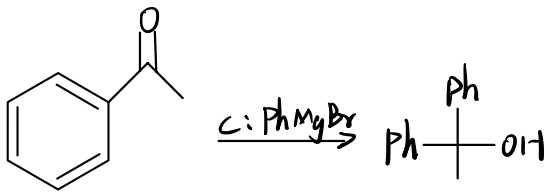
③ Resonance effect：NO_2 can delocalize negative charge through resonance increasing acidity

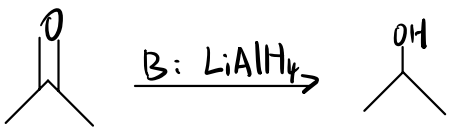
题五：carbonyl compound and alcohol

从两个O连接的C开始断  ， 

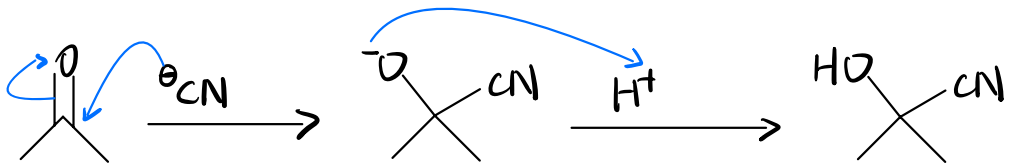
题六：reagents and mechanisms

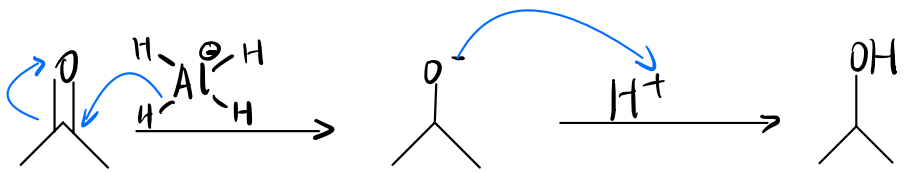


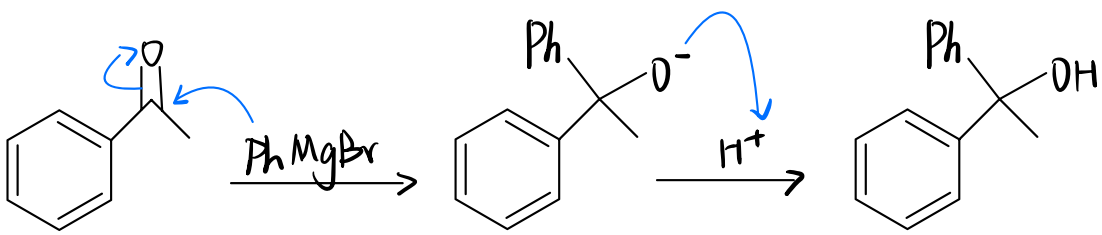




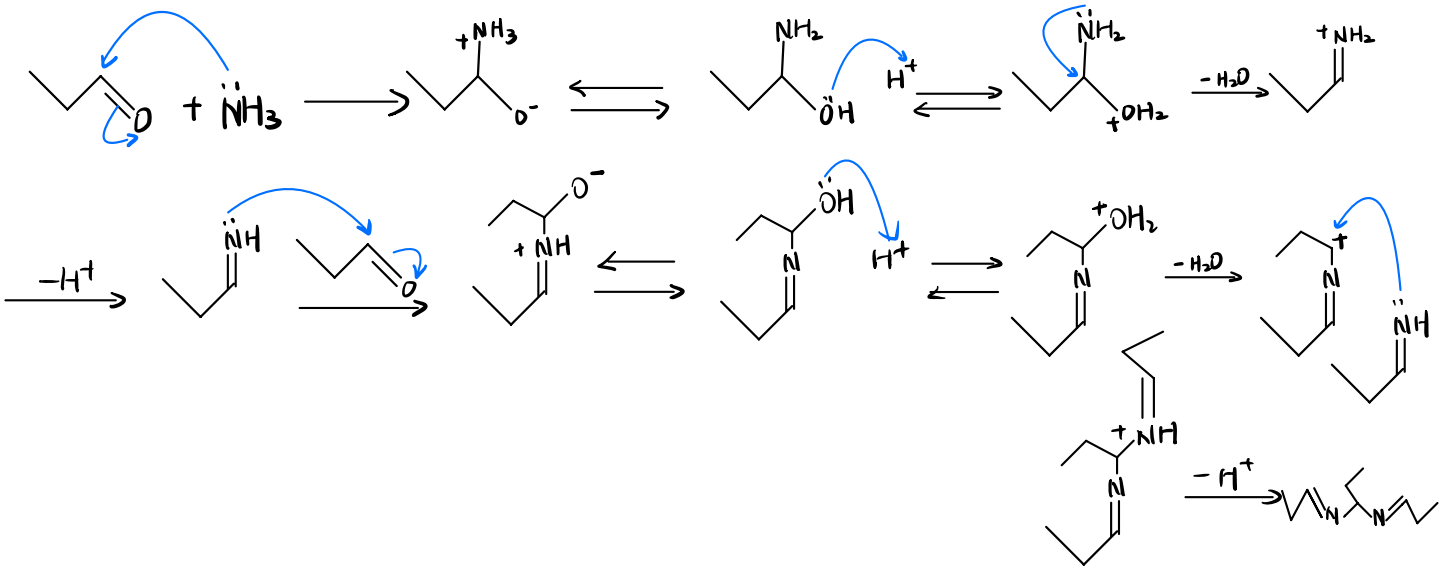
mechanisms：







题七：machanisms



題八：Ac_2O 